

Group name: VOIDBOUND WILL
Course: Object Oriented Programming

Team Members

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Introduction & Objectives

Introduction:

Our project is a simple parking management system.

It manages vehicles, drivers and parking slots.

It generates tickets and calculates parking fees.

Objectives:

To make parking management easier.

To reduce manual parking work.

To keep parking information organized.

Interface & Vehicle:

Ipayable Interface:

Defines the calculatefee() method for payment calculation.

Vehicle Abstract Class:

Stores common vehicle information like owner, brand, and number plate.

Displayvehicle() Virtual Method:

Displays common vehicle information and can be overridden.

Vehicle Types & Driver:

Car, Motorcycle, Cng:

Inherit the vehicle class and add their own information.

Override Method:

Each vehicle type overrides displayvehicle() and displayvehicletype().

Driver Class:

Stores driver name and phone number using constructors.

Parking Slot Class:

Parkingslot Class:

Represents each parking space in the parking system.

Fields:

Stores slot number, slot type, occupied status, and parked vehicle.

Static Field:

totalslot keeps track of the total number of parking slots.

Slot Operations:

(assigned) Method:

Assigns a vehicle to an available parking slot.

released() Method:

Removes the vehicle and makes the slot available again.

DisplayPSinfo() Method:

Displays the current parking slot information.

Ticket & Constructors:

parkingticket Class:

Stores ticket number, vehicle, slot, entry time, and exit time.

Constructor:

Creates a new parking ticket with vehicle and slot information.

Copy Constructor:

Creates a new ticket object by copying an existing ticket.

Fee & Operator Overloading:

calculatefee() Method:

Calculates the parking fee based on parking time.

Method Overloading:

calculatefee() and calculatefee(double) calculate fees using different rates.

Operator Overloading:

== and != compare two parking tickets by ticket number.

System Management:

parking system Class:

Manages drivers, vehicles, slots, tickets, and revenue.

registervehicle() and registereddriver():

Register vehicles and drivers in the system.

findavailableslot():

Finds a suitable empty slot for a vehicle.

Parking Operations :

ParkVehicle() Method:

Assigns a slot and generates a parking ticket for the vehicle.

exitvehicle() Method:

Calculates the fee, releases the slot, and updates revenue.

generatedailyreport() Method:

Shows total vehicles, drivers, slots, tickets, and revenue.

Object Creation & Parking:

Main() Method:

Creates objects of the parking system, drivers, and different vehicles.

Registration:

Registers drivers and vehicles into the parking system.

Parking:

Creates parking tickets and assigns suitable parking slots.

OOP Demonstration:

Copy Constructor:

Copies ticket1 into copiedTicket.

Method & Operator Overloading:

Demonstrates custom fee calculation and ticket comparison.

OOP Concepts:

The project demonstrates interface, abstraction, inheritance, overriding, constructors, static methods, and other required features.

Description & Conclusion:

- .The Smart Parking System manages vehicles, drivers, parking slots, tickets, fees, and daily reports using C# OOP concepts.
- .The project makes parking management simple and organized while demonstrating the required OOP features.

output:

```
... PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POLYGLOT NOTEBOOK >_

SYSTEM
VoidBound will
successfully register the driver
successfully register the driver
successfully registered vehicle
successfully registered vehicle
successfully registered vehicle

g system.c...

VEHICLE INFORMATION:
Vehicle owner:Arifa,vehicle brand:Toyota,Vehicle numberplate:1234
vehicletype:car,seats=5

Vehicle owner:munni,vehicle brand:Yamaha,Vehicle numberplate:5678
vehicletype:motorcycle,carrier=True

Vehicle owner:nusrat,vehicle brand:Bajaj,Vehicle numberplate:9999
vehicletype:CNG,seatnum=3

    avaiable slot:
slot:1,Type of vehicle in slot:car,currently:False,parkedvehicle:
slot:2,Type of vehicle in slot:motorcycle,currently:False,parkedvehicle:
slot:3,Type of vehicle in slot:cng,currently:False,parkedvehicle:
slot:4,Type of vehicle in slot:General,currently:False,parkedvehicle:

    avaiable slot:
slot:1,Type of vehicle in slot:car,currently:False,parkedvehicle:
slot:2,Type of vehicle in slot:motorcycle,currently:False,parkedvehicle:
slot:3,Type of vehicle in slot:cng,currently:False,parkedvehicle:
slot:4,Type of vehicle in slot:General,currently:False,parkedvehicle:

Vehicle parked successfully.

parkingticket
Ticket Number=1001,vehicle =1234,Vehicletype=car,Parking Slot:1,Entrytime=8/20/2026 6:17:28 AM,

Vehicle parked successfully.

parkingticket
Ticket Number=1002,vehicle =5678,Vehicletype=motorcycle,Parking Slot:2,Entrytime=8/20/2026 6:17:28 AM,

OPERATOR OVERLOADING :
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parking ticket

Ticket Number=1002, vehicle =5678, Vehicle type=motorcycle, Parking Slot:2, Entry time

OPERATOR OVERLOADING :

Original Ticket: 1001

Copied Ticket: 1001

METHOD OVERLOADING:

Normal Fee: 0 Tk

Custom Rate Fee: 100 Tk

Ticket marked as lost

lost ticket

lost ticket fee: 200 Tk

Parking slot released.

VEHICLE EXIT :

Vehicle: 1234

Slot: 1

Parking Fee: 0 Tk

Parking slot released.

Ticket marked as lost

DAILY PARKING REPORT :

Total registered Drivers: 2, Total Registered vehicles: 3, Total Parking Slots: 4

**Thank You For Your
Patients**